

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2020 Worked Solutions

January 2020 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 January 2020

January 2020 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. The table below shows corresponding values of x and y for $y = \log_2(2x)$

The values of y are given to 2 decimal places as appropriate.

x	2	5	8	11	14
y	2	3.32	4	4.46	4.81

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for $\int_2^{14} \log_2(2x) dx$, giving your answer to one decimal place. (3)

Using your answer to part (a) and making your method clear, estimate

- (b) (i) $\int_2^{14} \frac{\log_2(4x^2)}{5} dx$
(ii) $\int_2^{14} \log_2\left(\frac{2}{x}\right) dx$ (4)

(Total 7 marks)

Worked Solution - Question 1

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $- = - = - \int \int$

Final answer

$- = - = - \int \int$

Question 2

Binomial Expansion

2. One of the terms in the binomial expansion of $(3 + ax)^6$, where a is a constant, is $540x^4$

(a) Find the possible values of a .

(4)

(b) Hence find the term independent of x in the expansion of

$$\left(\frac{1}{81} + \frac{1}{x^6}\right)(3 + ax)^6$$

(3)

(Total 7 marks)

Worked Solution - Question 2

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: $4 4 2 a a = \text{therefore} =$
 $+/- (b) 6 6 1 3 81 a \times + = 17$

Final answer

$4 4 2 a a = \text{therefore} = +/- (b) 6 6 1 3 81 a \times + = 17$

Question 3

Polynomials

3.

$$f(x) = 6x^3 + 17x^2 + 4x - 12$$

- (a) Use the factor theorem to show that
- $(2x + 3)$
- is a factor of
- $f(x)$
- .

(2)

- (b) Hence, using algebra, write
- $f(x)$
- as a product of three linear factors.

(4)

- (c) Solve, for
- $\frac{\pi}{2} < \theta < \pi$
- , the equation

$$6 \tan^3 \theta + 17 \tan^2 \theta + 4 \tan \theta - 12 = 0$$

giving your answers to 3 significant figures.

(2)

(Total 8 marks)

Worked Solution - Question 3

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 4

4.

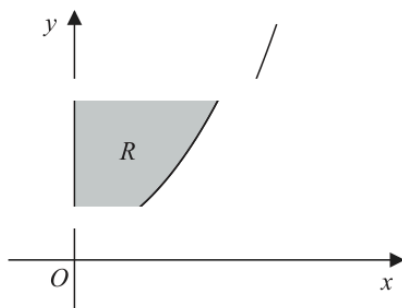


Figure 1

Figure 1 shows a sketch of the curve with equation

$$y = 2x^2 + 7 \quad x \geq 0$$

The finite region R , shown shaded in Figure 1, is bounded by the curve, the y -axis and the line with equation $y = 17$

Find the exact area of R .

(6)

(Total 6 marks)

Worked Solution - Question 4

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 5

Geometric Sequences

5.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

A colony of bees is being studied.

The number of bees in the colony at the start of the study was 30 000

Three years after the start of the study, the number of bees in the colony is 34 000

A model predicts that the number of bees in the colony will increase by $p\%$ each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

Assuming the model,

- (a) find the value of p , giving your answer to 2 decimal places.

(3)

According to the model, at the end of N years of study the number of bees in the colony exceeds 75 000

- (b) Find, showing all steps in your working, the smallest integer value of N .

(5)

(Total 8 marks)

Worked Solution - Question 5

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: [5]

Final answer

[5]

Question 6

Circles

6. The circle C has equation

$$x^2 + y^2 + 6x - 4y - 14 = 0$$

- (a) Find

- (i) the coordinates of the centre of C ,
 (ii) the exact radius of C .

(3)

The line with equation $y = k$, where k is a constant, is a tangent to C .

- (b) Find the possible values of k .

(2)

The line with equation $y = p$, where p is a negative constant, is a chord of C .

Given that the length of this chord is 4 units,

- (c) find the value of p .

(3)

(Total 8 marks)

Worked Solution - Question 6

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: = - - therefore 2 23 p = -

Final answer

= - - therefore 2 23 p = -

Question 7

Trigonometric Equations

7. (a) Show that the equation

$$8 \tan \theta = 3 \cos \theta$$

may be rewritten in the form

$$3 \sin^2 \theta + 8 \sin \theta - 3 = 0$$

(3)

- (b) Hence solve, for $0 \leq x \leq 90^\circ$, the equation

$$8 \tan 2x = 3 \cos 2x$$

giving your answers to 2 decimal places.

(4)

(Total 7 marks)

Worked Solution - Question 7

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: degrees degrees

Final answer

degrees degrees

Question 8

Sequences & Series

8. (i) An arithmetic series has first term a and common difference d .

Prove that the sum to n terms of this series is

$$\frac{n}{2}\{2a + (n-1)d\} \quad (3)$$

- (ii) A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_n = 5n + 3(-1)^n$$

Find the value of

(a) u_5 (1)

(b) $\sum_{n=1}^{59} u_n$ (3)

(Total 7 marks)

Worked Solution - Question 8

Topic group

1. Translate the sequence rule

Write the recurrence, sigma expression, or general term exactly from the question before simplifying.

2. Find the requested term or sum

Evaluate the sequence or series in order, then state the required term, total, or conclusion.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 9

Laws of Logarithms

9. (a) Sketch the curve with equation

$$y = 3 \times 4^x$$

showing the coordinates of any points of intersection with the coordinate axes.

(2)

The curve with equation $y = 6^{1-x}$ meets the curve with equation $y = 3 \times 4^x$ at the point P .

- (b) Show that the x coordinate of P is $\frac{\log_{10} 2}{\log_{10} 24}$

(5)

(Total 7 marks)

Worked Solution - Question 9

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: therefore therefore d

Final answer

therefore therefore d

Question 10

Applications of Differentiation

10. A curve C has equation

$$y = 4x^3 - 9x + \frac{k}{x} \quad x > 0$$

where k is a constant.

The point P with x coordinate $\frac{1}{2}$ lies on C .

Given that P is a stationary point of C ,

(a) show that $k = -\frac{3}{2}$ (4)

(b) Determine the nature of the stationary point at P , justifying your answer. (2)

The curve C has a second stationary point.

(c) Using algebra, find the x coordinate of this second stationary point. (4)

Blank

Q10

(Total 10 marks)

(Total 10 marks)

Worked Solution - Question 10

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: $2 \frac{1}{2} x^2 + 2x + 2$ therefore =
therefore = oe

Final answer

$2 \frac{1}{2} x^2 + 2x + 2$ therefore = therefore = oe